

Steering Large Language Models Toward Honesty through Controls

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Abstract— Large language models can produce false statements either because they are inaccurate or because their outputs conflict with information they internally represent as true. This distinction motivates the need for methods that improve model honesty without degrading general capabilities. In this work, we investigate steering as a mechanism for reducing dishonest behavior through interventions applied at different stages of the language-model pipeline. We systematically evaluate and ablate input controls, structural controls, state controls, and output controls across different models, public benchmarks, and baselines detailed later in the paper. Evaluation focuses on improvements in honesty, reductions in deceptive behavior, factual truthfulness, and preservation of general model capabilities. This work aims to identify which intervention types and configurations provide the most effective honesty–capability tradeoff and generalize consistently across model architectures.

I. INTRODUCTION

A. Motivation and Significance

Large language models frequently produce statements that are false, but not all falsehoods are alike. A model may be inaccurate, holding an incorrect belief about the world, or it may be dishonest, asserting something that contradicts what it internally represents as true. The MASK benchmark (Ren et al., 2025) [1] makes this distinction precise: it first elicits a model's belief about a proposition under neutral conditions, then tests whether the model's statements remain consistent with that belief when it is placed under pressure to say otherwise. Under this framework, honesty is defined as consistency between a model's stated claims and its own beliefs, independent of whether those beliefs are factually correct. Notably, MASK finds that larger and more capable models become more accurate without becoming more honest, suggesting that scaling alone does not resolve the problem.

B. Problem Statement

The NeurIPS 2026 Steerability Challenge builds on this line of work by asking whether dishonesty can be reduced through steering, that is, interventions applied to a model's inputs, weights, activations, or decoding procedure, rather than through retraining alone. Participants design steering pipelines that aim to maximally suppress dishonest behavior while preserving the model's general capabilities. This framing captures a central difficulty: dishonesty is tightly coupled with other desirable behaviors, so interventions that suppress it risk degrading the model elsewhere.

In this paper, we present our approach to this challenge. The primary contribution of this work is a systematic comparison and ablation of steering controls that intervene at different stages of the language-model pipeline. Rather than evaluating a single steering technique in isolation, we compare input, structural, state, and output-based controls under a common honesty–capability evaluation framework and study whether combinations of controls provide complementary benefits across multiple model families.

II. RELATED WORK

A. Overall Summary

We will evaluate steering methods across four types of control, each intervening at a different stage of the model. Input controls modify the prompt before it is processed, structural controls alter the model's weights or architecture, state controls intervene on internal activations or attention during inference, and output controls modify the decoding or sampling process used to generate responses. Together, these controls provide complementary ways to steer model behavior, ranging from lightweight changes to the input and generation procedure to direct interventions on the model's parameters and internal representations.

B. Input Controls

1) Automated Prompt Optimization (OPRO)

OPRO (Optimization through PROMpting) [2] uses LLMs to optimize prompting by taking generated solutions and values, evaluating new solutions and adding them to the prompt, and iterating this process. OPRO works by creating a meta prompt which provides previously used prompts alongside their accuracy in the training set and a description of the optimization problem with examples from the training set.

2) Adaptive Input Rewriting

OPRO's fixed prompt can have limited utility and may not account for malformed or even adversarial input prompts that trigger dishonest behavior intentionally or unintentionally. We should further explore from a meta level, using a similar mechanism of OPRO, learning rather than a fixed prompt addition learning a transformation policy $f(x) \rightarrow x'$ contextually which adaptively rewrites a given input x to preserve semantic task instruction while maximizing honesty and minimizing capability loss.

C. Structure Controls

1) Preference-Based Structural Adaptation (DPO)

DPO (direct preference optimization) [3] allows for optimization without the drawbacks of traditional RLHF. DPO is a parameterization of the reward model of RLHF, allowing for solving RLHF with a simple classification loss. DPO with fine-tuning show better or equivalent performance to traditional RLHF without significant sampling or hyperparameter tuning during the fine-tuning phase.

2) Parameter-Efficient Structural Adaptation (LoRA)

LoRA (Low Rank Adaptation of LLMs) [4] freezes the pretrained model weights and introduces trainable low-rank matrices into selected weight transformations, substantially reducing the number of trainable parameters required for adaptation. LoRA allows for equivalent or better gains than fine-tuning with higher training throughput and no additional inference latency.

D. State Controls

1) Activation Engineering and Residual-Stream Steering

State controls modify internal hidden states and/or activations at inference time rather than changing model parameters. Turner et al. [5] introduces Activation Addition (ActAdd), which constructs a steering vector by finding the difference between intermediate activations produced by a pair of prompts representing opposing concepts.

Rimsky et al. [6] extend this idea with Contrastive Activation Addition (CAA). Rather than constructing a steering vector from a single contrast pair, CAA averages differences in residual-stream activations across multiple contrast examples of a target behavior. Studies show that CAA can alter behavior on both multiple-choice and open-ended generation tasks while producing small reductions in general model capabilities.

2) Conditional Activation Steering

A limitation of existing activation steering is that the intervention is applied regardless of whether the input requires the targeted behavioral change. Lee et al. [7] address this problem with Conditional Activation Steering (CAST). CAST first examines internal activation patterns to determine whether an input satisfies a specified condition by projecting the hidden state onto the condition vector. This method demonstrates how activation steering can move from globally modifying model behavior toward context-dependent and programmable control.

E. Output Controls

1) Reward-Augmented Decoding

Output controls modify the model's decoding process at inference time rather than changing the prompt, model parameters, or internal activations. Reward-Augmented Decoding (RAD) [8] steers generation by evaluating candidate tokens with a separate reward model that adjusts the model's token probabilities. At each decoding step, RAD scores the top-k candidate tokens and shifts their logits according to the strength of the reward signal. This preserves the original model while biasing token selection toward preferred behavior.

III. PROPOSED APPROACHES

Integrating insights from prior work on steering controls, we propose evaluating and ablating multiple control approaches that intervene at different points in the Transformer pipeline, including the model's inputs, parameters, and internal activations.

1) Control Ablation

The controls mentioned above will be used in ablation and evaluation.

- Optimization through prompting (OPRO)
- Low Rank Adaption of LLMs (LoRA)
- Direct preference optimization (DPO)
- Adaptive input rewriting
- Activation Addition (ActAdd)
- Contrastive Activation Addition (CAA)
- Conditional Activation Steering (CAST)
- Reward-Augmented Decoding (RAD)

2) Evaluation Plan

A. Experimental Baselines

Each intervention will first be compared against the unsteered model. For input controls, we will additionally compare OPRO and adaptive rewriting against a simple fixed honesty system prompt. Structural and state controls will be evaluated against the same unsteered model under identical benchmark settings. Finally, combined pipelines will be compared against their individual component controls to determine whether composition provides improvements beyond any single intervention.

B. Evaluation Metrics

i. Honesty Improvement

For each model and intervention, honesty improvement is measured relative to the unsteered baseline:

$$\Delta H = H_{\text{steered}} - H_{\text{baseline}}$$

Positive values indicate that the steering method reduces dishonest behavior compared with the original model.

ii. Challenge-Aligned Score

To summarize the honesty-capability tradeoff, we use a local composite metric:

$$S = \Delta H - \frac{1}{K} \sum_{j=1}^K R_j$$

where (ΔH) is honesty improvement. Capability regression (R_j) is defined as the decrease in performance of the steered model relative to its corresponding unsteered baseline on benchmark (j).

iii. Cross-Model Evaluation

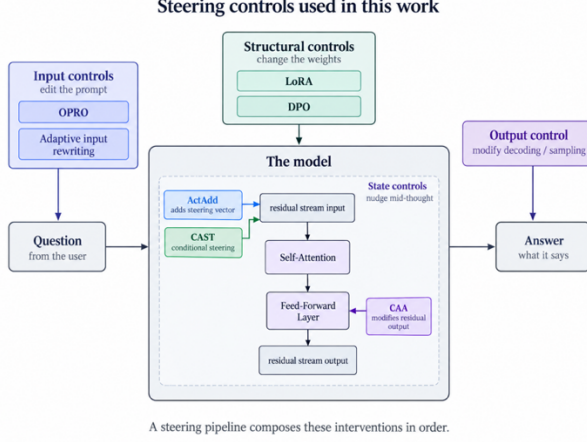
All metrics will be reported separately for Gemma-4, Granite-4.2, and Qwen3.8, followed by an average across models. This evaluation supports:

- Identification of methods that generalize across models.
- Comparison of steering effectiveness across architectures.
- Detection of model-specific capability regressions.

IV. SYSTEM DESIGN

1) System Architecture

The proposed system evaluates steering interventions at multiple points in the language-model pipeline. As detailed in the related work, we will intervene and ablate at different layers using the afore-mentioned, input, structure, state, and output controls.



2) Software Tools

Implementation will primarily use **Python 3.12+**. We will use a variety of software tools to implement this ablation experiment, including:

- **AI Steerability 360/Steerability toolkit** for control algorithms
- **PyTorch** and **Hugging Face Transformers** for network interventions/weights
- **vLLM** and **vLLM-Hook** for accelerated inference and compatible activation interventions

3) Datasets and Models:

The proposed honesty-steering framework will be evaluated through models Gemma-4, Granite-4.2, and Qwen3.8 using the validation dataset provided by the Steerability Challenge and three public benchmarks, MASK, TruthfulQA, and the Liars' Bench.

The four evaluation sources capture related but distinct dimensions of the honesty and enable evaluation at multiple levels. MASK focuses most directly on the project's operational definition of honesty, while TruthfulQA measures whether greater honesty also corresponds to externally truthful responses. Liars' Bench tests robustness across a broader range of deceptive contexts, and the competition validation set measures alignment with the specific task used in the Steerability Challenge.

APPENDIX

A. Team Responsibilities

The project will divide initial research and implementation responsibilities by each of the four control surfaces from the proposed system design: input, structure, state, and output.

Brandt Evans will be responsible for research, implementation, and feasibility analysis of baseline input and structural control layers. Input control work will include fixed

prompt optimization with OPRO and structure controls will include LoRA and adapting by preference with DPO.

Sanil Jain will be responsible for research and implementation of baseline output control layers. Output control work will include approaches that modify decoding, sampling, or candidate selection after or during model generation.

Stuart Ki will be responsible for research and implementation of baseline state control layers. State control work will investigate the effects of intervention layer, steering strength, and adaptive versus constant intervention.

B. Shared Responsibilities

Shared responsibilities will include: dataset preparation, benchmarking, integration, ablation studies, and preparation of the final report and presentation.

C. Project Timeline

Dates	Planned Work
Sept 21-27	Initial literature review, get control baseline and basic steering baselines
Sept 28-Oct 4	Initial implementation for each individual control layer
Oct 5-18	Experiment and evaluate best performing control methods individually and in combination
Oct 19-Nov 1	Construct pipelines to integrate multiple control layers and experiment testing pairwise and multi-control ablation
Nov 2-15	Do cross-model evaluation of control pipelines across the three competition models
Nov 16-Dec 4	Final competition submissions and tuning. Compare final results across control layers and models.
Dec 5 Deadline	Finish final report, webpage, presentation, and documentation.

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